Bag-of-Words trick

- Can you guess what this is about:
  - per is salary hour £5,594 Neymar’s

    Neymar’s salary per hour is £5,594
  - obesity French is of full cause and fat fries

    French fries is full of fat and cause obesity

- **Main idea**: Re-ordering doesn’t destroy the topic
  - individual words are “building blocks”
  - “bag” of words: a “composition” of “meanings”

Simplest: Bag-of-Words trick

- Most search engines use BOW
  - treat documents and queries as bags of words
- A “bag” is a set with repetitions
  - match = “degree of overlap” between ***d***, ***q***

- **Retrieval models**
  - statistical models (functions): usually use words as features
  - decide which documents most likely to be relevant
- BOW makes these models tractable (and also effective!)

What's the Simplest IR System?

- Given a collection of documents and a "free text" query
- How can we get some search results in a simple way?

- ***grep-like***: a "sequential scan"
- Simple but …
  - very inefficient
- Is it effective?

How can we make it more effective AND efficient?

Boolean Retrieval Model

- **Queries**: Users express queries as a ***Boolean expression***
  - AND, OR, NOT
  - Can be arbitrarily nested
- Ex. query: ***information*** *AND* ***retrieval*** *AND NOT* ***technology***

- **Documents**: Views each document as a ***"bag" of words***

- Return only documents that satisfy the Boolean query.

Boolean Retrieval Model

- Any given query divides the collection into two sets:
  - retrieved (matching)
  - not-retrieved (not matching)
- Returns a ***set*** of documents that "**exactly**" satisfy the query (Boolean expression)
  - Called "**Exact-Match**" retrieval
- Used?
  - Many search systems still in-use are Boolean
  - e.g., Email, library catalog, Mac OS X Spotlight, legal search

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"Term-Document Incidence Matrix" shows ...

- for each term, the documents that have it
- for each document, the terms that appear in it
- both

Boolean retrieval always find relevant documents

- Yes, always!
- Sometimes
- Never!

Bigger Collections …

- Consider N = 1 million documents, each with about 1000 words.
- Say there are M = 500K *distinct* terms among these.
- 500K x 1M matrix has half-a-trillion 0's and 1's.

- But it has no more than one billion 1's. ?
  - matrix is extremely sparse.

What's a better representation?